



The Sound of Music

>>The Web Audio API lets the internet talk to you beyond videos and music downloads> by Koustav Ray

Introduction

The internet has primarily been silent. Sound on the other hand is a powerful tool, almost as powerful as vision and at times even more. Leaving aside the regular audio or video streaming websites like SoundCloud or YouTube, the internet generally prefers to avoid tuning into the auditory aspect. The main reason for this was the complex and heavy Flash objects that were required to generate even simple sounds. With HTML5 the <audio> and <video> tags enabled better integration of pre-recorded sounds and videos; however, the original problem of creating dynamic sound at runtime still remained. Thus the Web Audio API in HTML5 came to the rescue, and made it possible to design web applications and games where different kinds of sounds ranging from a single tone to complex ones like drum beats could be generated on the fly. Web Audio therefore can now replace all the flash based sound generators and make webpages lighter and cleaner. The monotony of silent webpages is slowly but surely being broken with aesthetic use of sound effects.

Why audio?

Audio conveys meaning, can draw attention of the user to particular events (like an incoming call, notification, alert etc.) or

even create the ambience as desired by the UI designers. In fact, audio being used in User Interfaces is not uncommon at all. Most mobile applications take the help of sound to denote different user interactions like a touch, slide or swipe. Games have been using sound extensively since a long time, even in their menus and navigational links to provide feedback to



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the end users. Desktop applications like Skype, Yahoo Messenger, MS-Outlook etc. use sound effects extensively as well. In fact Operating Systems starting from Windows to UNIX based systems have also used sound to assist in user experience. Many

of these sounds have come to be associated with these applications in an inseparable way, so much so that some of the tunes provide identity to these applications; e.g.: the trademark sound of Startup & Shutdown in Microsoft's Windows XP, the jingle of a skype call, the intro of EA Sports, or even the jingle of erst-while video games like Super Mario. On the web too, sounds

*Coding Matters



*Chrome Samples

>>Web Audio samples from Google showing Web Audio API's full potential.

<http://dgit.in/WAudSamples>



*PowerShell on Linux

>>Microsoft open sources Powershell and makes it available for Linux.

<http://dgit.in/PwrShLOS>



*Intel RealSense tools

>>At the 2016 Intel Developer Forum, a number of tools for RealSense technology announced.

<http://dgit.in/IntelRS>